GROUP G12 · FOR IRTAZA KAZMI (23I-6001)

Paper Build Specification

Exactly what the Overleaf manuscript must contain: every section, every number, every reference. Nothing here needs to be looked up elsewhere.

00 Paper identity

Title	Thermally Mandated Offloading: Why Quantization Gains Vanish Under Sustained Edge LLM Inference
Authors	Faheem (23i-0728), Irtaza Kazmi (23i-6001) — FAST-NUCES Islamabad
Group / area	G12 · Edge–Cloud Intelligence
Overleaf name	`[Grp12_23i-0728_23i-6001_thermally_mandated_offloading]`
Instructor access	`basharat.hussain.v@isb.nu.edu.pk` as Editor
Repository	`github.com/MFaheemS/Cloud-Project`
Source files	`paper/main.tex`, `paper/references.bib` — upload both to Overleaf

If the Overleaf project was created from the instructor's template, do not overwrite its `main.tex`. Paste the section bodies into the template's matching sections and upload `references.bib` only.

01 Section structure and status

This is the required order (Section 14 of the assignment guide). Word counts are what is already in `paper/main.tex`.

#	SECTION	WORDS	STATUS	OWNER
1	Abstract	~130	placeholder — rewrite last	Faheem
2	Introduction	48	written, could grow	Faheem
3	Problem and Motivation	101	written	Faheem
4	Research Questions and Objectives	116	written	Faheem
5	**Literature Review**	158	expand to 900–1,200	**Irtaza**
6	**Research Gap**	29	expand to ~200	**Irtaza**
7	Methodology	731	complete — do not edit	Faheem
8	Experimental Setup	142	complete — do not edit	Faheem
9	Results	447	complete — do not edit	Faheem
10	Discussion	0	empty heading	together
11	Limitations	0	empty heading	together
12	Future Work	0	empty heading	together
13	Conclusion	0	empty heading	together
14	References	—	grows with §05	**Irtaza**
15	Authors Profile	0	short bio each	both

> Your edits are confined to **Literature Review**, **Research Gap** and **references.bib**. Sections 7–9 are finished and tied to the numbers below; changing them breaks the cross-references.

02 Everything the paper claims — all results

These are every measured number in the project. You will need them to write the gap, and to check nothing you write contradicts them.

A. BASELINE REPRODUCTION (VS HUSOM ET AL., TABLE 8)

200 prompts × 5 repeats = 1,000 generations per cell, scored with the authors' own scripts.

MODEL	DATASET	PAPER	OURS	DIFF	P
qwen2.5-0.5b-q4_K_M	CommonsenseQA	0.32	0.291	−0.029	0.420
qwen2.5-1.5b-q4_K_M	CommonsenseQA	0.62	0.593	−0.027	0.474
qwen2.5-0.5b-q4_K_M	TruthfulQA	0.26	0.275	+0.015	0.660
qwen2.5-1.5b-q4_K_M	TruthfulQA	0.24	0.249	+0.009	0.786

All four consistent; two-proportion z-test, no significant difference.

B. NEW MEASUREMENTS THE PAPER DOES NOT CONTAIN

Husom et al. report no q8_0 or fp16 row for the 1.5B model: neither fits on their 4 GB Raspberry Pi.

MODEL	COMMONSENSEQA	TRUTHFULQA
qwen2.5-1.5b-q4_K_M	0.593	0.249
qwen2.5-1.5b-q8_0	0.698	0.239
qwen2.5-1.5b-fp16	0.705	0.245

CommonsenseQA gains **10.5 points** from q4 to q8, then plateaus (+0.7 to fp16). TruthfulQA is flat, so the effect is task-dependent.

C. SUSTAINED-LOAD BEHAVIOUR (RTX 3050, 15 MIN CONTINUOUS)

CONFIGURATION	PEAK TOK/S	SUSTAINED TOK/S	LOSS	STEADY CLOCK
0.5B Q4	243.6	179.9	26.2%	1181 MHz
1.5B Q4	115.8	66.2	42.9%	839 MHz
1.5B Q8	95.4	66.8	30.0%	832 MHz
1.5B FP16 (CPU-offloaded)	25.3	24.6	2.9%	—

Key finding: Q4 leads Q8 by 21% at cold start (115.8 vs 95.4) but they converge to the same floor (66.2 vs 66.8). The speed advantage of aggressive quantization does not survive sustained load; its accuracy cost does.

D. THERMAL TIME CONSTANTS

QUANTITY	1.5B Q4	1.5B Q8
Temperature range under load	61.6 → 88.8 °C	56.2 → 88.4 °C
`tau_heat` (temperature)	25 s	30 s
`tau_cool` (temperature)	105 s	101 s
`tau_heat` (throughput)	239 s	269 s
Temperature saturates faster than throughput by	9.5×	8.9×

Second key finding: temperature reaches equilibrium roughly nine times faster than throughput degrades. The device looks thermally stable for the entire period in which most performance is lost, so temperature is a useless control signal. `SwThermalSlowdown` was active in 96.7% of telemetry samples.

E. PROPOSED ESTIMATOR AND ABLATION

The method fits `tok/s(t) = a + b·exp(−t/tau)` to a measurement window and reports the asymptote `a` instead of the window mean.

ARM	WHAT IT DOES	MAPE	WORST	COST
baseline	mean of first 60 s	51.1%	75.0%	60 s
+A	measure 300 s, last-60 s mean	18.1%	27.7%	300 s
+B	extrapolate from first 60 s	17.9%	39.2%	60 s
+A+B	extrapolate from first 300 s	10.5%	23.3%	300 s

B alone matches A's accuracy at one-fifth the measurement cost, so the gain comes from the model rather than from measuring longer.

CONFIG	ACTUAL	BASELINE	+A	+B	+A+B
0.5b-q4	179.9	243.6	188.8	175.4	179.9
1.5b-q4	66.2	115.8	84.5	58.3	81.6
1.5b-q8	66.8	95.4	81.2	93.0	61.3
1.5b-fp16	24.6	25.3	22.1	24.7	17.8

F. HEADROOM ANALYSIS (HOW MUCH ROOM IS LEFT)

WINDOW (S)	30	45	60	90	120	180	240	300	450	600
MAPE %	49.1	45.2	17.9	94.9	112.9	38.4	19.9	10.5	13.1	11.2

- Error is **non-monotonic**; at 90–120 s the estimator is worse than the baseline because the unconstrained fit returns physically impossible negative asymptotes (−42.9 and −61.4 tok/s).
- **Oracle** (same form fitted to the full run) reaches 3.8% mean error vs 10.5% deployed → **6.7 points recoverable**.
- **Double-exponential** fits better on all three runs; worst config improves 6.7% → 3.6%. Power-law is far worse.
- **Physical constraints** (0 < a ≤ min observed, b ≥ 0) help but do not fix it: mean across windows 41.3% → 31.9%, worst 192.8% → 100.0%.

G. ENVIRONMENT (ALREADY IN EXPERIMENTAL SETUP)

```
Device     NVIDIA RTX 3050 Laptop GPU, 4 GB, driver 581.83
Host       Windows 11 build 26200, Python 3.13.5
Serving    Ollama 0.24.0 (llama.cpp, GGUF)
Telemetry  NVML via nvidia-ml-py, 20 Hz
Models     Qwen2.5 0.5B / 1.5B Instruct; Q4_K_M, Q8_0, FP16
Datasets   CommonsenseQA 200, TruthfulQA MC 200
Protocol   --repeat 5, temperature 0; 15 min load + 15 min recovery, cold start <60 C
Upstream   MELODI d55dae9, data repo c11fe13
```

H. REPRODUCIBILITY PROBLEMS FOUND IN THE BASELINE'S CODE

1. No requirements file; `scikit-learn`, `rouge_score`, `python-Levenshtein`, `nltk` undocumented.

2. `signal.SIGALRM` is Unix-only — crashes on Windows.
3. Printing raw model output crashes under cp1252 when a model emits CJK text; one run died at 501/1000.
4. **Output opens in append mode**, so a re-run silently concatenates. Produced a 1,501-line file that the evaluator scored without complaint.
5. Energy measured with an external hardware meter on a Pi — not comparable to our NVML GPU sampling.
6. Model tags unpinned, so exact weights cannot be guaranteed.

Items 2–4 mean the artifacts have never been run on Windows. Item 4 is the serious one: results that look valid and are silently wrong.

---

03 Literature Review — full specification

Target **900–1,200 words** across four themed subsections, plus the comparison table. All 19 references are already in `paper/references.bib`, verified against the arXiv API and Crossref. **You do not need to find more papers.**

> Not a list of summaries. Each subsection groups papers by what they share, then states what they collectively miss. The missing things become the Research Gap, so the gap is proven rather than asserted.

5.1 EDGE LLM MEASUREMENT STUDIES — ~250 WORDS

CITE	WHAT IT DOES	LIMITATION TO DRAW OUT
`tummalapalli2026edge`	Sustained-load benchmark of Qwen2.5-1.5B Q4 on Pi5+NPU, two phones, RTX 4050 laptop	one model at one quantization level; measurement only, proposes no policy
`renney2026cloudedge`	Benchmarks LLM inference on hardware-accelerated single-board computers; reports throughput density and energy per million tokens	short benchmark runs; no sustained or thermal axis
`taherin2026hydra`	Phase-aware characterisation across edge SoC generations, backends and quantization levels	characterisation only; no estimator or deployment policy
`xiao2024pockets`	Performance study of LLMs on commodity mobile devices	mobile only; quantization–thermal interaction not isolated

Closing sentence must say: these studies measure devices in a cold or briefly warmed state, and none reports whether the figure they publish is sustainable.

5.2 ENERGY AND QUANTIZATION TRADE-OFFS — ~300 WORDS

CITE	WHAT IT DOES	LIMITATION TO DRAW OUT
husom2025sustainable	**Our baseline.** 28 quantized Ollama models on a Raspberry Pi 4 (4 GB), five datasets, hardware power meter	short runs; no sustained load; cannot host 1.5B above q4 on 4 GB
fadelargerich2026wattcounts	Energy benchmark, 50 LLMs across 10 GPUs, batch and server scenarios	FP16 only (quantization named as future work); steady state, no time axis
stuhlmann2025bench360	Local LLM benchmark across backends, quantization formats, tasks and scenarios	configs assume 24 GB VRAM; no sustained scenario; no GGUF backend, so it cannot run on real consumer GPUs
fadelargerich2026wattgpu	Predicts inference power and latency on unseen GPUs and LLMs	predicts across *hardware*, not across *time under load*
liu2026energytotoken	Position paper: inference should be evaluated as energy-to-token production	argues for better metrics but specifies no sustained-load protocol
liu2025flexquant	Dynamic precision switching driven by perplexity and KL divergence	switches on output-quality signals, never on thermal state

Closing sentence must say: quantization is evaluated for speed, energy and accuracy, always assuming the measured rate is the steady-state rate; none verifies that assumption.

Use `fadelargerich2026wattgpu` to position our estimator: both predict, but along different axes. Say so explicitly.

5.3 THERMAL AND POWER-AWARE MANAGEMENT — ~300 WORDS

CITE	WHAT IT DOES	LIMITATION TO DRAW OUT
stojkovic2025tapas	Thermal- and power-aware scheduling for LLM inference in cloud datacentres	needs facility cooling and power-capping controls consumer hardware does not expose
liu2025greenllm	SLO-aware dynamic frequency scaling, separate control for prefill and decode	datacentre GPUs; requires direct frequency control
zou2026enerinfer	Energy-aware on-device inference with DVFS on mobile-class hardware	single device; no quantization ladder; no placement decision
lu2026therme	**Closest related work.** Predictive management of shared thermal headroom for sustained LLM inference on edge SoCs, on vLLM	predicts *temperature headroom*, not sustained throughput; no quantization axis; no placement — see §06
benoitcattin2020thermal	Long-term CNN inference on a Raspberry Pi 4B under thermal throttling, with and without active cooling	CNNs, not LLMs; CPU only — but establishes the phenomenon predates LLM serving

Closing sentence must say: thermal-aware systems either target datacentres with control knobs consumer hardware lacks, or manage temperature at runtime; none asks whether the *published performance figure* for a configuration is achievable under sustained load.

5.4 EDGE–CLOUD PLACEMENT AND OFFLOADING — ~250 WORDS

CITE	WHAT IT DOES	LIMITATION TO DRAW OUT
yang2024perllm	Personalised inference scheduling with edge–cloud collaboration	decides from a service rate measured in advance
younesi2025splitwise	Collaborative edge–cloud inference, splitting adaptively with Lyapunov-assisted DRL as bandwidth varies	adapts to *network* variation; treats per-device compute capability as stable
yang2025moaoff	Modality-aware offloading for multimodal LLM inference	offloads on modality and workload, never on device condition
akgul2025cosense	Cost- and uncertainty-aware cloud–edge cooperation	optimises cost and uncertainty; device throughput assumed fixed

Closing sentence must say: every placement system assumes a fixed per-device service rate, and our measurements show that rate is wrong by up to 75 % within minutes.

`akgul2025cosense`: confirm this is a credible paper before citing. If not, drop it — the subsection stands on the other three.

COMPARISON TABLE — INCLUDE IN THE PAPER

One row per paper, same columns as above plus Metrics. Build it before writing prose, then write each subsection by reading down the **Limitation** column. Numbers taken from abstracts must be checked in the paper itself before they go in the table.

---

04 Research Gap — full specification

Target ~200 words. Four points, each tied to a measurement from §02, then one closing claim.

1. **No sustained-load dimension.** Studies in 5.1–5.2 measure short runs. Our data: 43 % throughput loss over 15 minutes, and Q4's 21 % advantage over Q8 disappears entirely at equilibrium.
2. **Temperature carries no information during the loss.** It saturates ~9× faster than throughput degrades (25 s vs 239 s), so systems in 5.3 that monitor temperature are blind through the window where performance is lost.
3. **Placement uses stale rates.** Systems in 5.4 decide from a pre-measured service rate; ours is wrong by up to 75 %.
4. **A hardware ceiling in the baseline.** Husom et al. could not host 1.5B above q4 on a 4 GB Pi, so their guidance for that size is incomplete — and the unmeasured region contains a 10.5-point accuracy gain.

Closing claim: no existing work measures how quantization level affects a device's ability to *sustain* performance, nor connects that to the placement decision.

---

05 All 19 references

Every entry below is in `paper/references.bib` and was verified on 2026-10-10 against the arXiv API or Crossref: title, full author list, year, DOI. Cite with `\cite{key}`.

BIBTEX KEY	REFERENCE	THEME
husom2025sustainable	Husom, Goknil, Astekin, Shar, Kåsen, Sen, Mithassel, Soylu. Sustainable LLM Inference for Edge AI. *ACM TIOT* 6(4), 2025. doi:10.1145/3767742 ← **baseline**	5.2
tummalapalli2026edge	Tummalapalli, Arayakandy, Pal, Kundan. LLM Inference at the Edge. arXiv:2603.23640, 2026	5.1
renney2026cloudedge	Renney, Trad, Mattarock, Evetts, Wood. Cloud to Edge: Benchmarking LLM Inference on Hardware-Accelerated Single-Board Computers. arXiv:2604.24785, 2026	5.1
taherin2026hydra	Taherin, Taghipour Anvari, Amante, Chen, Noroian, Feric, Bohm Agostini, Zhao, Cano, Ren, Wang, Kaeli. Hydra: Phase-Aware Workload Characterization of LLM Inference. arXiv:2608.25053, 2026	5.1
xiao2024pockets	Xiao, Huang, Chen, Tian. Understanding Large Language Models in Your Pockets. arXiv:2410.03613, 2024	5.1
fadelargerich2026wattcounts	Fadel Argerich, Fürst, Patiño-Martínez. Watt Counts. arXiv:2604.09048, 2026	5.2
stuhlmann2025bench360	Stuhlmann, Fadel Argerich, Fürst. Bench360: Benchmarking Local LLM Inference from 360 Degrees. arXiv:2511.16682, 2025	5.2
fadelargerich2026wattgpu	Fadel Argerich, Fürst, Patiño-Martínez. WattGPU: Predicting Inference Power and Latency on Unseen GPUs and LLMs. arXiv:2607.02391, 2026	5.2
liu2026energytotoken	Liu, Yuan, Tang, Dong, Zhao, Wang, Li, Chu. Position: LLM Inference Should Be Evaluated as Energy-to-Token Production. arXiv:2605.11733, 2026	5.2
liu2025flexquant	Liu, Wang, Xia, Zhao, Zhao, Li, Liu, Jiang, Guan. FlexQuant. arXiv:2506.12024, 2025	5.2
stojkovic2025tapas	Stojkovic, Zhang, Goiri, Choukse, Qiu, Fonseca, Torrellas, Bianchini. TAPAS. ASPLOS '25. doi:10.1145/3676641.3716025	5.3
liu2025greenllm	Liu, Huang, Zapater, Atienza. GreenLLM: SLO-Aware Dynamic Frequency Scaling. arXiv:2508.16449, 2025	5.3
zou2026enerinfer	Zou, Liu, Sun, Mascherin, Roy, Liu, Peng, Jia, Chen. EnerInfer. arXiv:2606.23001, 2026	5.3
lu2026therme	Lu, Wang, Liu. ThermE: Predictive Management of Shared Thermal Headroom for Sustained LLM Inference on Thermally Constrained Edge SoCs. arXiv:2610.00267, 2026 ← **closest work**	5.3
benoitcattin2020thermal	Benoit-Cattin, Velasco-Montero, Fernández-Berni. Impact of Thermal Throttling on Long-Term Visual Inference in a CPU-based Edge Device. arXiv:2010.06291, 2020	5.3
yang2024perllm	Yang, Yang, Zhao, Guo, He, Ji. PerLLM. arXiv:2405.14636, 2024	5.4
younesi2025splitwise	Younesi, Shabrang Maryan, Oustad, Najafabadi Samani, Ansari, Fahringer. Splitwise. 2025. doi:10.1145/3773274.3774267	5.4
yang2025moaoff	Yang, Guo, Hu, Zhao, Zhang, Zhao, Ji. MoA-Off. arXiv:2509.16995, 2025	5.4
akgul2025cosense	Akgul, Eplik, Rojas, Abdullah, van der Merwe. CoSense-LLM. arXiv:2510.19670, 2025 *(verify credibility)*	5.4

RULES

- Open and read every paper you cite — at minimum abstract, method, conclusion. The bibliographic data is verified; the *claims* you make about each paper are not.
- Any number you quote from a paper must be checked in that paper, not taken from a search summary. Two repository URLs handed to us in this project turned out not to exist.
- Never copy sentences. Paraphrase and cite. No promotional language. Every factual claim carries a citation.

06 Positioning against ThermE — do not skip

`lu2026therme` is the nearest work to ours and appeared recently. It must be cited in 5.3 *and* distinguished in the Research Gap, or a reviewer will raise it for us.

	THERME	THIS WORK
Predicts	thermal headroom (temperature)	sustained throughput (tok/s)
When	at runtime, continuously	once, from a short cold-start window
Purpose	schedule actions during serving	correct the figure a benchmark publishes
Quantization	not varied	the central variable
Placement	not addressed	the motivating decision
Platform	edge SoCs, vLLM	consumer GPU, llama.cpp/GGUF

One sentence in 5.3 and one in the gap is enough. Suggested framing: ThermE manages headroom while serving; we ask whether the performance figure a configuration is *advertised* with is attainable at all under sustained load, and show the answer depends on quantization level.

ThermE also notes that vendor governors react only near throttling thresholds. That is adjacent to our finding that temperature saturates long before throughput stops falling. Acknowledge it rather than presenting ours as wholly new — our measurement is of a different quantity and is quantified (9×).

07 How the marks are earned

CRITERION	MARKS	WHAT SATISFIES IT HERE
Quality / relevance of literature	5	19 references, 2024–2026, peer-reviewed where available (ACM TIOT, ASPLOS), every one verified against the published record
Literature synthesis	5	four themes, each grouping papers by a shared limitation; the comparison table; **no paper-by-paper summaries**
Research gap	5	four points, each supported by one of our measurements, plus explicit positioning against ThermE
RQ & objectives	3	already written — check the gap you write leads into them without contradiction
Expected contribution	2	already stated in Methodology and Results
Paper quality (final)	4	compiles cleanly, consistent terminology, all tables captioned and referenced, no broken citations

Marks are lost most often on synthesis, not on paper count. Fifteen well-grouped papers beat twenty summarised one by one.

08 Overleaf and working rules

1. **Upload** `paper/main.tex` and `paper/references.bib` from the repo. If the project was created from the instructor's template, do *not* overwrite its `main.tex` — paste section bodies into the template's sections and upload `references.bib` only.
2. **Project name:** `[Grp12_23i-0728_23i-6001_thermally_mandated_offloading]`
3. **Instructor as Editor:** `basharat.hussain.v@isb.nu.edu.pk`, access kept for the whole project.
4. **One project, both authors.** No personal copies.
5. **Only edit** Literature Review, Research Gap and `references.bib`. Sections 7–9 are finished and cross-referenced.
6. **Commit as you go** — git history is evidence of individual contribution.
7. **Add a row to** `CONTRIBUTIONS.md` each session. Every entry is currently Faheem's; either of us can be questioned on any part at the defence.

BEFORE HANDING BACK, CHECK

- Document compiles in Overleaf with no errors and no `??` in place of citations.

- Every new BibTeX entry is cited; every citation resolves.
- Each of 5.1–5.4 ends with its required closing sentence.
- The comparison table is captioned and referenced from the text.
- Nothing you wrote contradicts the numbers in §02.
- The abstract's last line is still the GitHub link.
